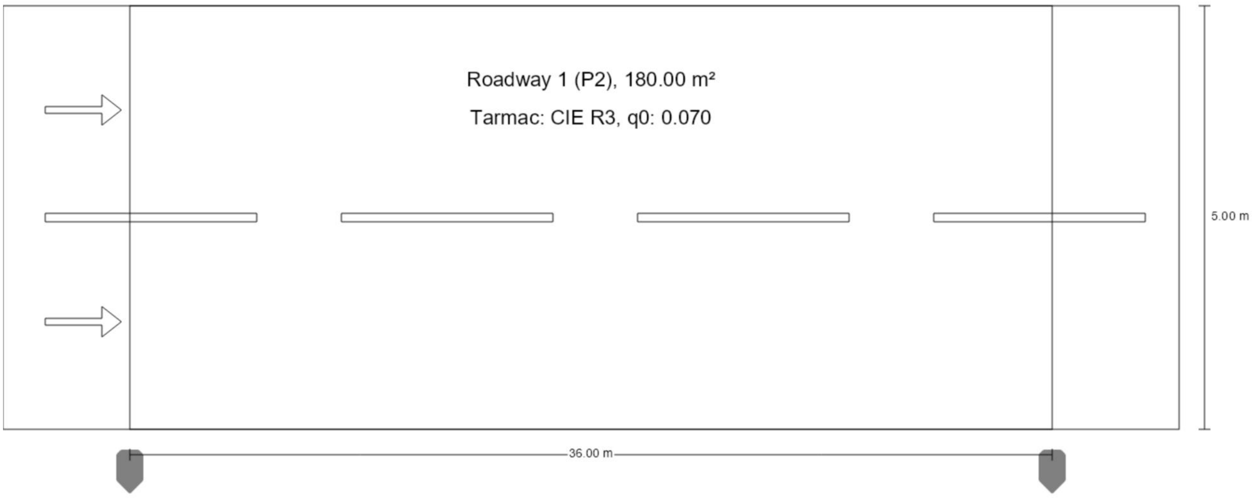
Table of Contents

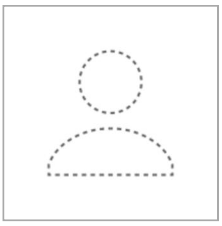
H : 5 D 36 Dim 100% CIE R3 Q0 0.07

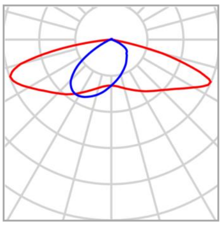
Summary (according to EN 13201:2015)



H : 5 D 36 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)





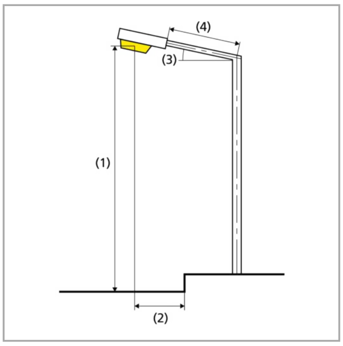
Article name	SA-2A01-060
Fitting	1x

P	40.1 W
Φ_{Lamp}	6829 lm
$\Phi_{Luminaire}$	6829 lm
η	100.00 %

H : 5 D 36 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)



Pole distance	36.000 m
(1) Light spot height	5.000 m
(2) Light point overhang	-0.500 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 100.0 %, 40.1 W
Wattage / route	1122.8 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.3
MF	0.80

H : 5 D 36 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.12 lx	[10.00 - 15.00] lx	✓
	E_{min}	2.29 lx	≥ 2.00 lx	✓

Results for energy efficiency indicators

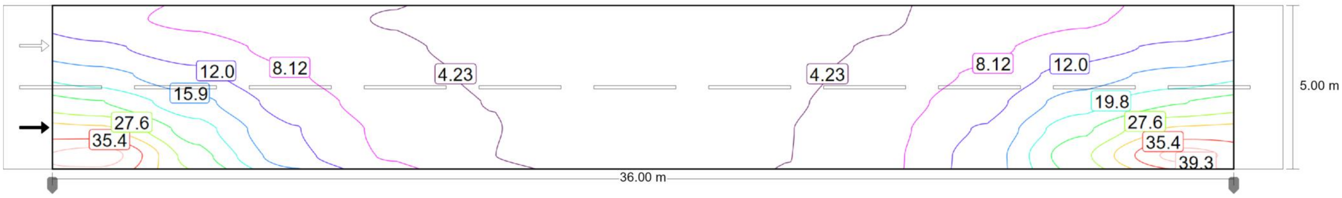
	Symbol	Calculated	Energy Consumption
H : 5 D 36 Dim 100% CIE R3 Q0 0.07	D_p	0.022 W/lx*m²	–
SA-2A01-060 (single side bottom)	D_e	0.9 kWh/m² yr	160.4 kWh/yr

H : 5 D 36 Dim 100% CIE R3 Q0 0.07

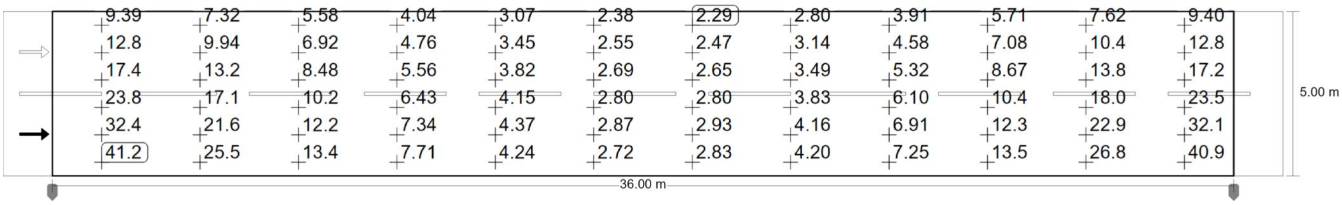
Roadway 1 (P2)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.12 lx	[10.00 - 15.00] lx	✓
	E_{min}	2.29 lx	≥ 2.00 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

m	1.500	4.500	7.500	10.500	13.500	16.500	19.500	22.500	25.500	28.500	31.500	34.500
4.583	9.39	7.32	5.58	4.04	3.07	2.38	2.29	2.80	3.91	5.71	7.62	9.40
3.750	12.81	9.94	6.92	4.76	3.45	2.55	2.47	3.14	4.58	7.08	10.36	12.76
2.917	17.40	13.21	8.48	5.56	3.82	2.69	2.65	3.49	5.32	8.67	13.83	17.22
2.083	23.81	17.15	10.24	6.43	4.15	2.80	2.80	3.83	6.10	10.44	18.05	23.49
1.250	32.38	21.63	12.16	7.34	4.37	2.87	2.93	4.16	6.91	12.34	22.90	32.12
0.417	41.20	25.51	13.42	7.71	4.24	2.72	2.83	4.20	7.25	13.52	26.84	40.86

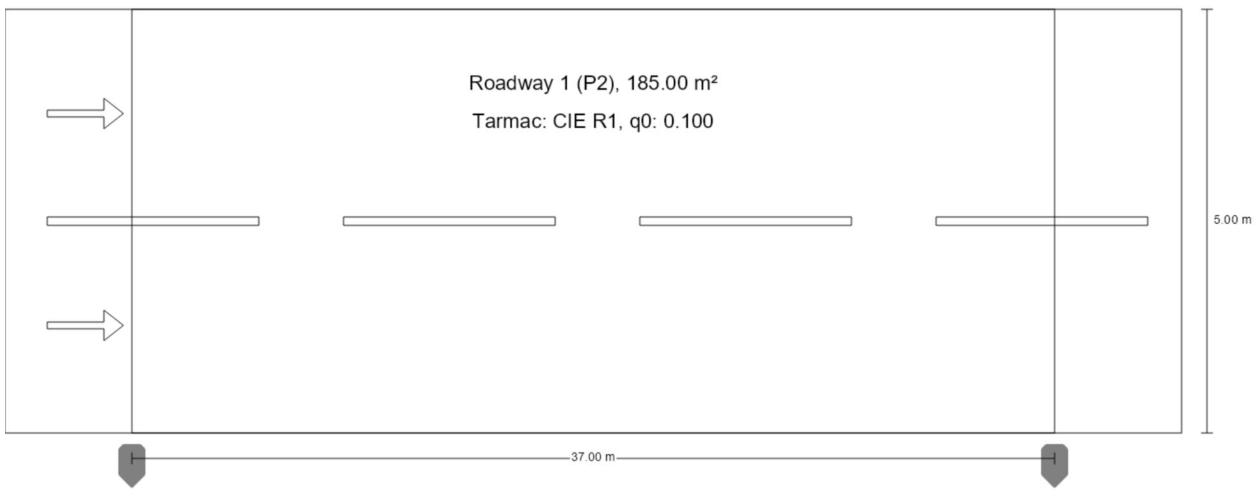
Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	10.1 lx	2.29 lx	41.2 lx	0.23	0.06

DIALux

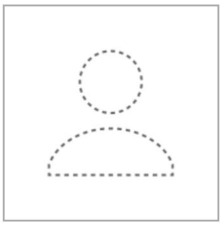
H : 5 D 37 Dim 100% CIE R1 Q0 0.10

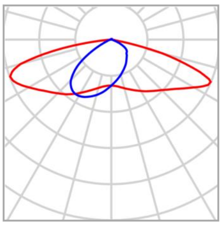
Summary (according to EN 13201:2015)



H : 5 D 37 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)





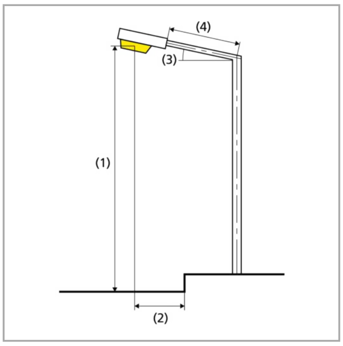
Article name	SA-2A01-060
Fitting	1x

P	40.1 W
Φ_{Lamp}	6829 lm
$\Phi_{Luminaire}$	6829 lm
η	100.00 %

H : 5 D 37 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)



Pole distance	37.000 m
(1) Light spot height	5.000 m
(2) Light point overhang	-0.393 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 100.0 %, 40.1 W
Wattage / route	1082.7 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.3
MF	0.80

DIALux

H : 5 D 37 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.10 lx	[10.00 - 15.00] lx	✓
	E_{min}	2.07 lx	$\geq$ 2.00 lx	✓

Results for energy efficiency indicators

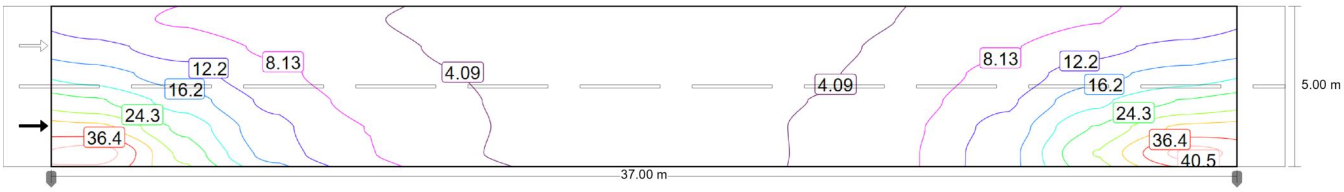
	Symbol	Calculated	Energy Consumption
H : 5 D 37 Dim 100% CIE R1 Q0 0.10	D_p	0.021 W/lx*m^2	–
SA-2A01-060 (single side bottom)	D_e	0.9 kWh/m^2 yr	160.4 kWh/yr

H : 5 D 37 Dim 100% CIE R1 Q0 0.10

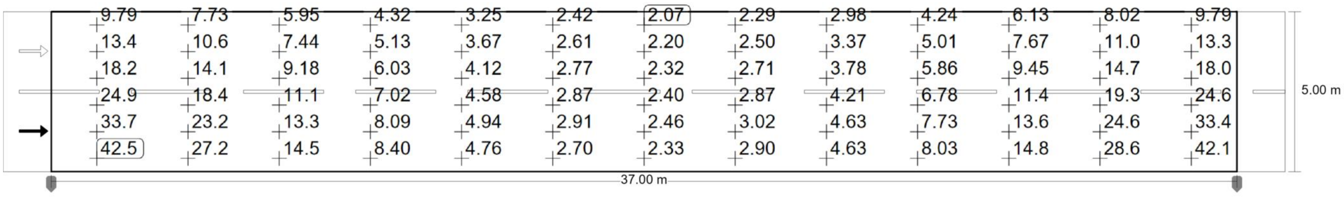
Roadway 1 (P2)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.10 lx	[10.00 - 15.00] lx	✓
	E_{min}	2.07 lx	≥ 2.00 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

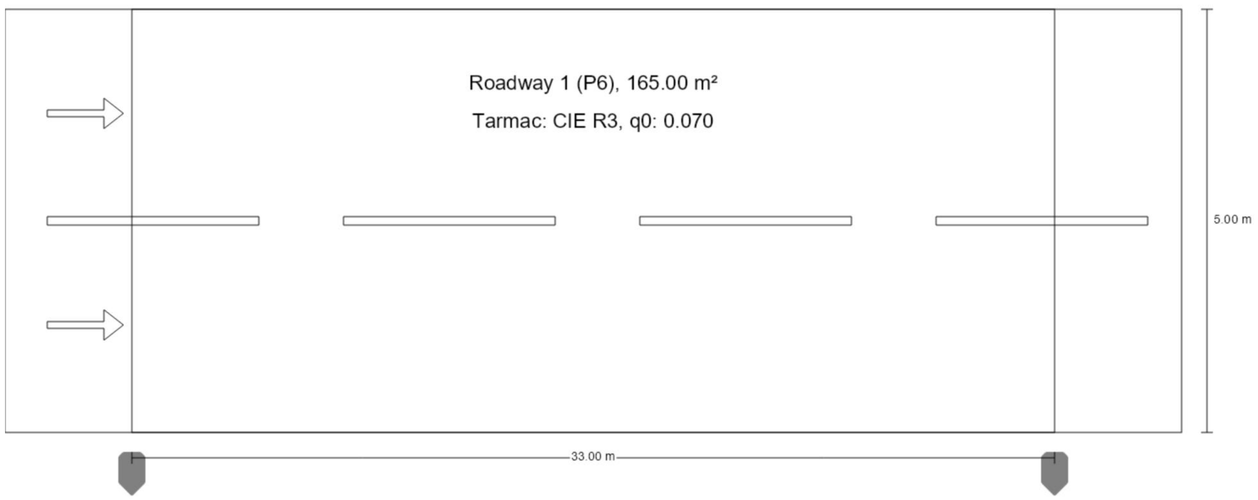
m	1.423	4.269	7.115	9.962	12.808	15.654	18.500	21.346	24.192	27.038	29.885	32.731	35.577
4.583	9.79	7.73	5.95	4.32	3.25	2.42	2.07	2.29	2.98	4.24	6.13	8.02	9.79
3.750	13.37	10.55	7.44	5.13	3.67	2.61	2.20	2.50	3.37	5.01	7.67	10.98	13.31
2.917	18.20	14.10	9.18	6.03	4.12	2.77	2.32	2.71	3.78	5.86	9.45	14.74	17.99
2.083	24.89	18.37	11.14	7.02	4.58	2.87	2.40	2.87	4.21	6.78	11.45	19.32	24.56
1.250	33.66	23.23	13.29	8.09	4.94	2.91	2.46	3.02	4.63	7.73	13.60	24.59	33.38
0.417	42.51	27.21	14.54	8.40	4.76	2.70	2.33	2.90	4.63	8.03	14.79	28.58	42.10

Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	10.1 lx	2.07 lx	42.5 lx	0.20	0.05

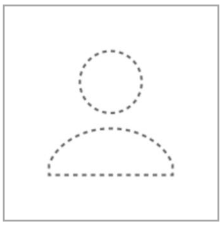
H : 6 D 33 Dim 25% CIE R3 Q0 0.07

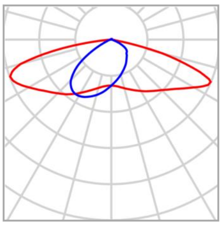
Summary (according to EN 13201:2015)



H : 6 D 33 Dim 25% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)





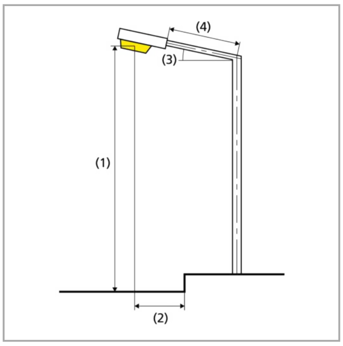
Article name	SA-2A01-060
Fitting	user-defined

P	10.0 W
Φ_{Lamp}	1707 lm
$\Phi_{Luminaire}$	1707 lm
η	100.00 %

H : 6 D 33 Dim 25% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)



Pole distance	33.000 m
(1) Light spot height	6.000 m
(2) Light point overhang	-0.493 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 25.0 %, 2.5 W
Wattage / route	300.0 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.5
MF	0.80

DIALux

H : 6 D 33 Dim 25% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	2.57 lx	[2.00 - 3.00] lx	✓
	E_{min}	0.93 lx	≥ 0.40 lx	✓

Results for energy efficiency indicators

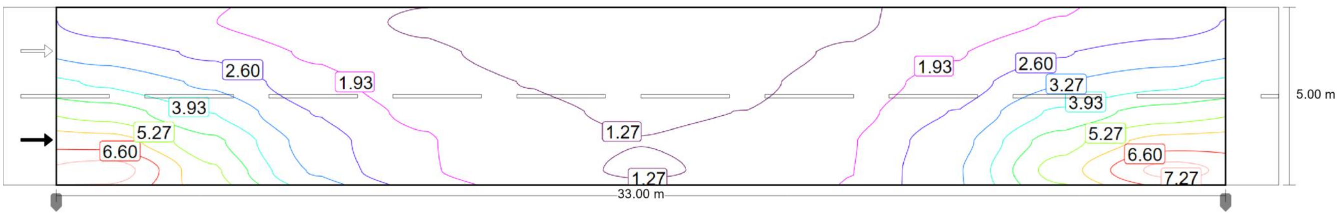
	Symbol	Calculated	Energy Consumption
H : 6 D 33 Dim 25% CIE R3 Q0 0.07	D_p	0.024 W/lx*m²	–
SA-2A01-060 (single side bottom)	D_e	0.1 kWh/m² yr	10.0 kWh/yr

H : 6 D 33 Dim 25% CIE R3 Q0 0.07

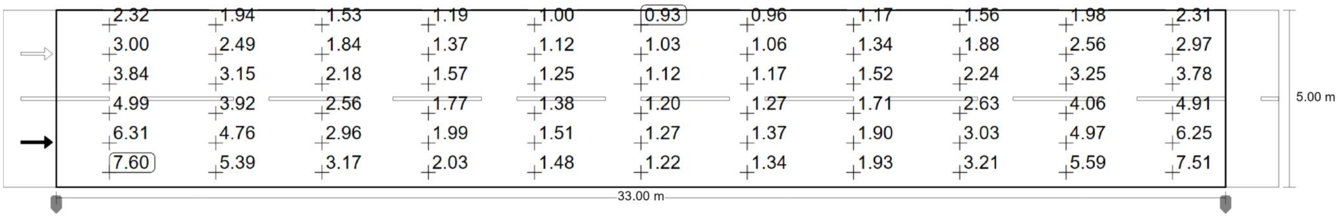
Roadway 1 (P6)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	2.57 lx	[2.00 - 3.00] lx	✓
	E_{min}	0.93 lx	≥ 0.40 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

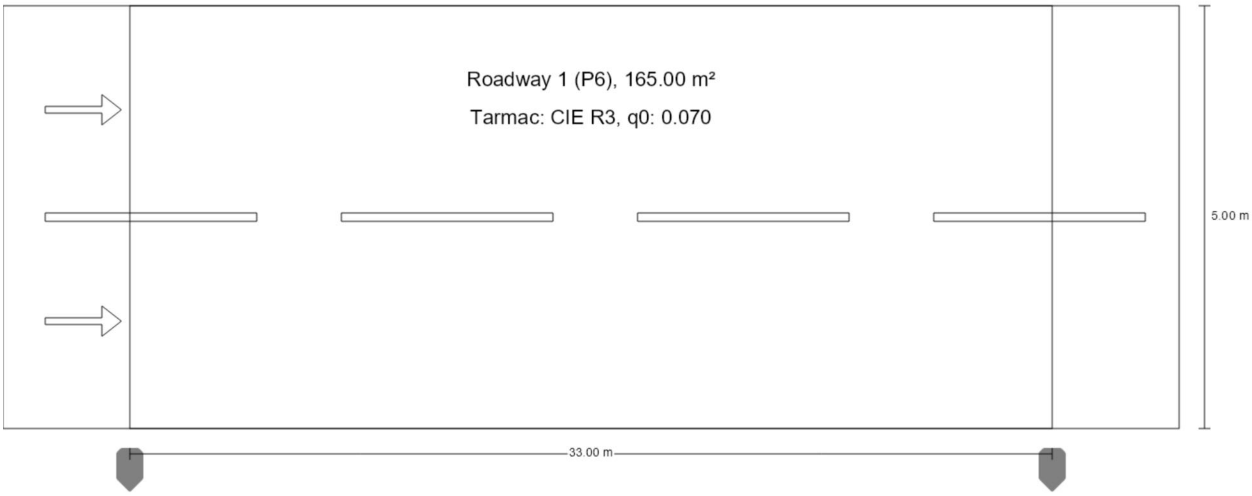
m	1.500	4.500	7.500	10.500	13.500	16.500	19.500	22.500	25.500	28.500	31.500
4.583	2.32	1.94	1.53	1.19	1.00	0.93	0.96	1.17	1.56	1.98	2.31
3.750	3.00	2.49	1.84	1.37	1.12	1.03	1.06	1.34	1.88	2.56	2.97
2.917	3.84	3.15	2.18	1.57	1.25	1.12	1.17	1.52	2.24	3.25	3.78
2.083	4.99	3.92	2.56	1.77	1.38	1.20	1.27	1.71	2.63	4.06	4.91
1.250	6.31	4.76	2.96	1.99	1.51	1.27	1.37	1.90	3.03	4.97	6.25
0.417	7.60	5.39	3.17	2.03	1.48	1.22	1.34	1.93	3.21	5.59	7.51

Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	2.57 lx	0.93 lx	7.60 lx	0.36	0.12

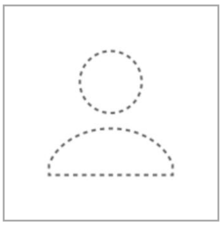
H : 6 D 33 Dim 100% CIE R3 Q0 0.07

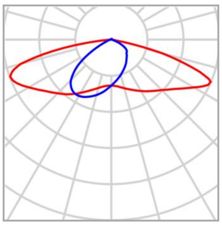
Summary (according to EN 13201:2015)



H : 6 D 33 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)





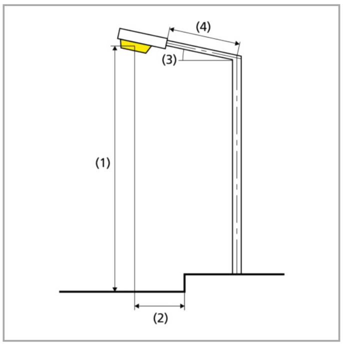
Article name	SA-2A01-060
Fitting	1x

P	40.1 W
Φ_{Lamp}	6829 lm
$\Phi_{Luminaire}$	6829 lm
η	100.00 %

H : 6 D 33 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)



Pole distance	33.000 m
(1) Light spot height	6.000 m
(2) Light point overhang	-0.493 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 20.0 %, 8.0 W 0 h: 20.0 %, 8.0 W
Wattage / route	1203.0 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.3
MF	0.80

H : 6 D 33 Dim 100% CIE R3 Q0 0.07

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	10.29 lx	[2.00 - 3.00] lx	✗
	E_{min}	3.73 lx	≥ 0.40 lx	✓

Results for energy efficiency indicators

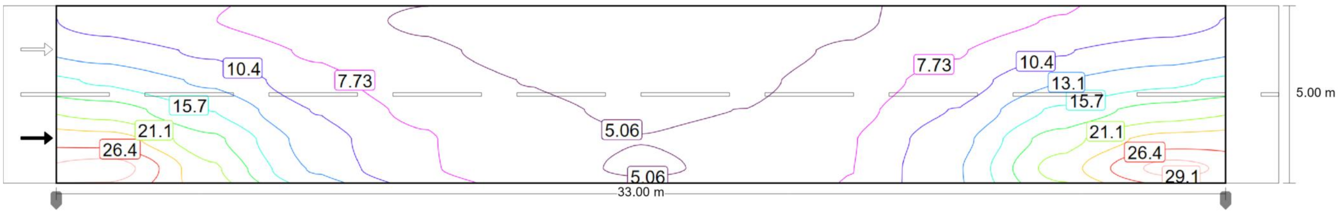
	Symbol	Calculated	Energy Consumption
H : 6 D 33 Dim 100% CIE R3 Q0 0.07	D_p	0.024 W/lx*m²	–
SA-2A01-060 (single side bottom)	D_e	0.2 kWh/m² yr	32.1 kWh/yr

H : 6 D 33 Dim 100% CIE R3 Q0 0.07

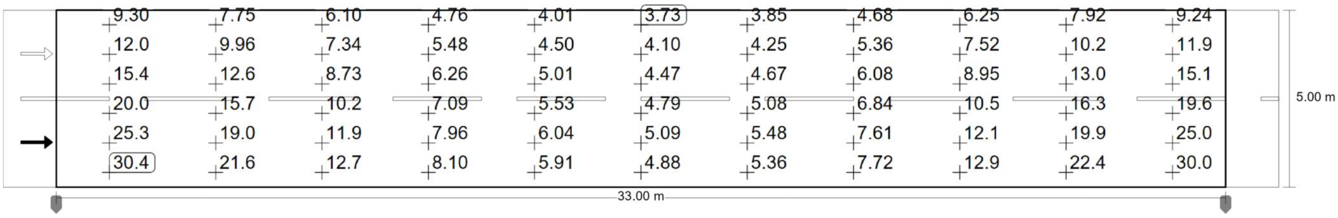
Roadway 1 (P6)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	10.29 lx	[2.00 - 3.00] lx	✗
	E_{min}	3.73 lx	≥ 0.40 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

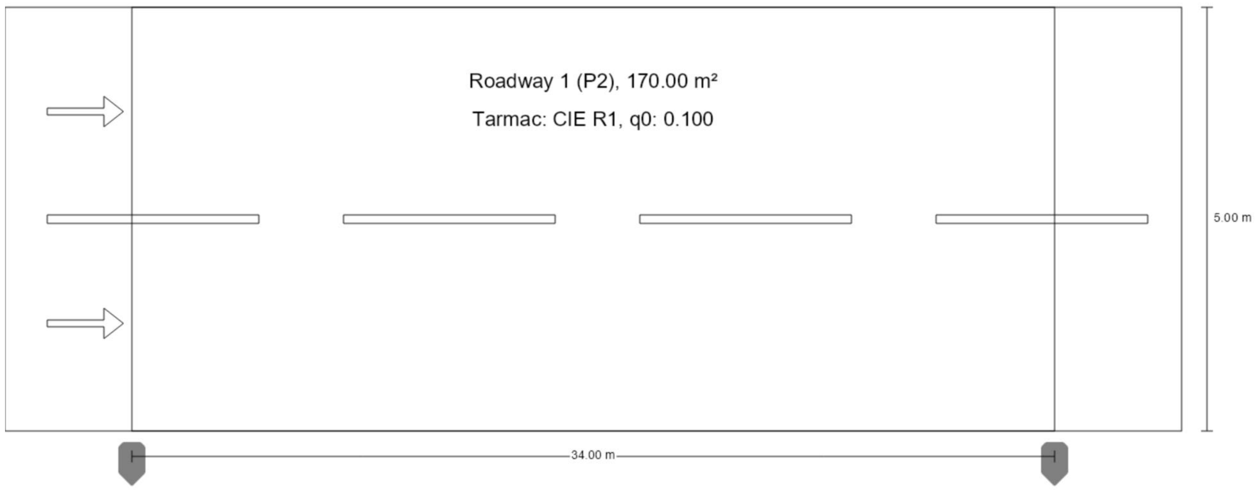
m	1.500	4.500	7.500	10.500	13.500	16.500	19.500	22.500	25.500	28.500	31.500
4.583	9.30	7.75	6.10	4.76	4.01	3.73	3.85	4.68	6.25	7.92	9.24
3.750	12.01	9.96	7.34	5.48	4.50	4.10	4.25	5.36	7.52	10.23	11.88
2.917	15.37	12.62	8.73	6.26	5.01	4.47	4.67	6.08	8.95	13.01	15.12
2.083	19.96	15.68	10.25	7.09	5.53	4.79	5.08	6.84	10.50	16.26	19.64
1.250	25.25	19.05	11.85	7.96	6.04	5.09	5.48	7.61	12.13	19.90	24.99
0.417	30.42	21.57	12.67	8.10	5.91	4.88	5.36	7.72	12.86	22.36	30.05

Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	10.3 lx	3.73 lx	30.4 lx	0.36	0.12

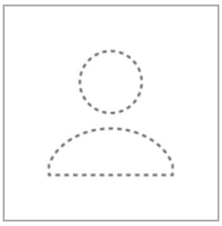
H : 6 D 34 Dim 100% CIE R1 Q0 0.10

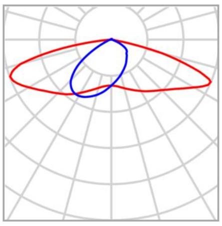
Summary (according to EN 13201:2015)



H : 6 D 34 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)





Article name	SA-2A01-060
Fitting	1x

P	40.1 W
Φ_{Lamp}	6829 lm
$\Phi_{Luminaire}$	6829 lm
η	100.00 %

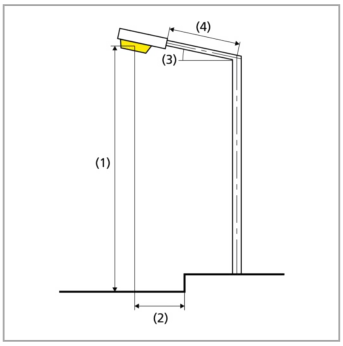
DIALux

H : 6 D 34 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)

Pole distance	34.000 m
(1) Light spot height	6.000 m
(2) Light point overhang	-0.393 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 100.0 %, 40.1 W
Wattage / route	1162.9 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.3
MF	0.80



H : 6 D 34 Dim 100% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.19 lx	[10.00 - 15.00] lx	✓
	E_{min}	3.47 lx	≥ 2.00 lx	✓

Results for energy efficiency indicators

	Symbol	Calculated	Energy Consumption
H : 6 D 34 Dim 100% CIE R1 Q0 0.10	D_p	0.023 W/lx*m²	–
SA-2A01-060 (single side bottom)	D_e	0.9 kWh/m² yr	160.4 kWh/yr

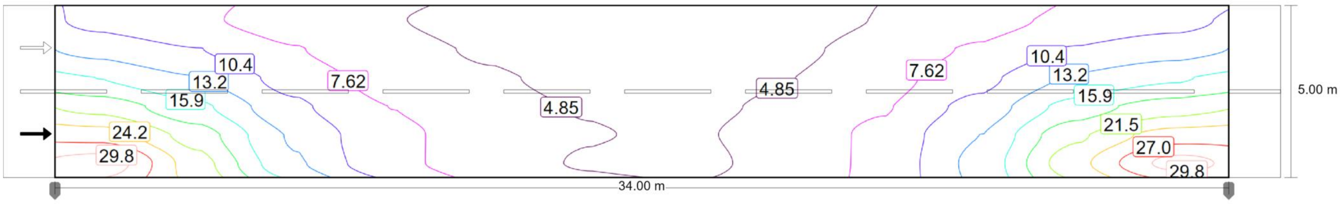
DIALux

H : 6 D 34 Dim 100% CIE R1 Q0 0.10

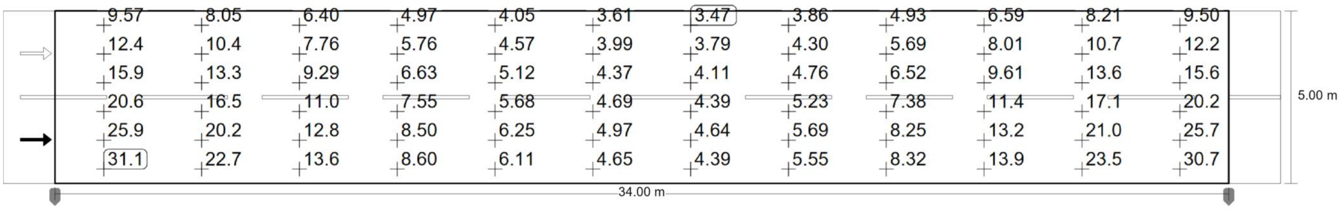
Roadway 1 (P2)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P2)	E_{av}	10.19 lx	[10.00 - 15.00] lx	✓
	E_{min}	3.47 lx	≥ 2.00 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

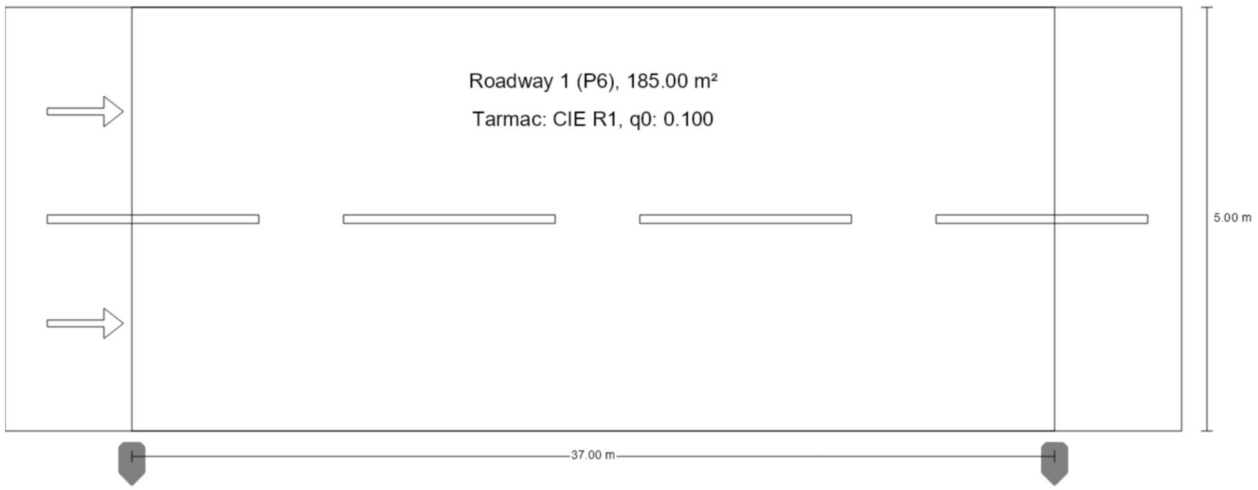
m	1.417	4.250	7.083	9.917	12.750	15.583	18.417	21.250	24.083	26.917	29.750	32.583
4.583	9.57	8.05	6.40	4.97	4.05	3.61	3.47	3.86	4.93	6.59	8.21	9.50
3.750	12.39	10.42	7.76	5.76	4.57	3.99	3.79	4.30	5.69	8.01	10.67	12.25
2.917	15.86	13.26	9.29	6.63	5.12	4.37	4.11	4.76	6.52	9.61	13.64	15.60
2.083	20.57	16.55	10.98	7.55	5.68	4.69	4.39	5.23	7.38	11.35	17.11	20.23
1.250	25.94	20.16	12.77	8.50	6.25	4.97	4.64	5.69	8.25	13.19	21.01	25.67
0.417	31.14	22.73	13.62	8.60	6.11	4.65	4.39	5.55	8.32	13.93	23.47	30.74

Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	10.2 lx	3.47 lx	31.1 lx	0.34	0.11

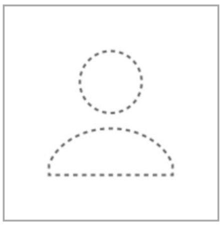
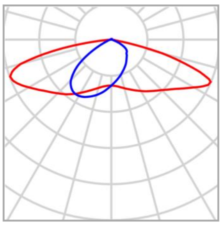
H : 6 D 37 Dim 30% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)



H : 6 D 37 Dim 30% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

Article name	SA-2A01-060
Fitting	user-defined

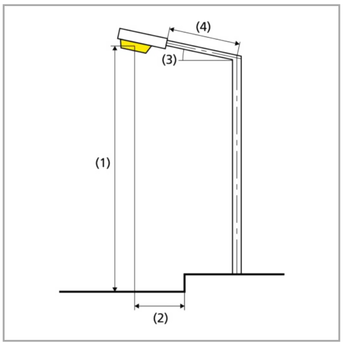
P	12.0 W
Φ_{Lamp}	2029 lm
$\Phi_{Luminaire}$	2029 lm
η	100.00 %

H : 6 D 37 Dim 30% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

SA-2A01-060 (single side bottom)

Pole distance	37.000 m
(1) Light spot height	6.000 m
(2) Light point overhang	-0.393 m
(3) Boom inclination	15.0°
(4) Boom length	0.000 m
Annual operating hours	4000 h: 30.0 %, 3.6 W
Wattage / route	324.0 W/km
ULR / ULOR	0.02 / 0.01
Max. luminous intensities Any direction forming the specified angle from the downward vertical, with the luminaire installed for use.	≥ 70°: 443 cd/klm ≥ 80°: 252 cd/klm ≥ 90°: 39.7 cd/klm
Luminous intensity class The luminous intensity values in [cd/klm] for calculation of the luminous intensity class refer to the luminaire luminous flux according to EN 13201:2015.	–
Glare index class	D.5
MF	0.80



H : 6 D 37 Dim 30% CIE R1 Q0 0.10

Summary (according to EN 13201:2015)

Results for valuation fields

A maintenance factor of 0.80 was used for calculating for the installation.

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	2.78 lx	[2.00 - 3.00] lx	✓
	E_{min}	0.82 lx	$\geq$ 0.40 lx	✓

Results for energy efficiency indicators

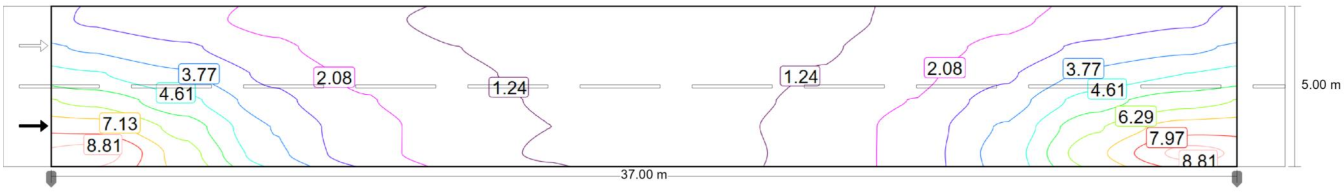
	Symbol	Calculated	Energy Consumption
H : 6 D 37 Dim 30% CIE R1 Q0 0.10	D_p	0.023 W/lx*m²	–
SA-2A01-060 (single side bottom)	D_e	0.1 kWh/m² yr	14.4 kWh/yr

H : 6 D 37 Dim 30% CIE R1 Q0 0.10

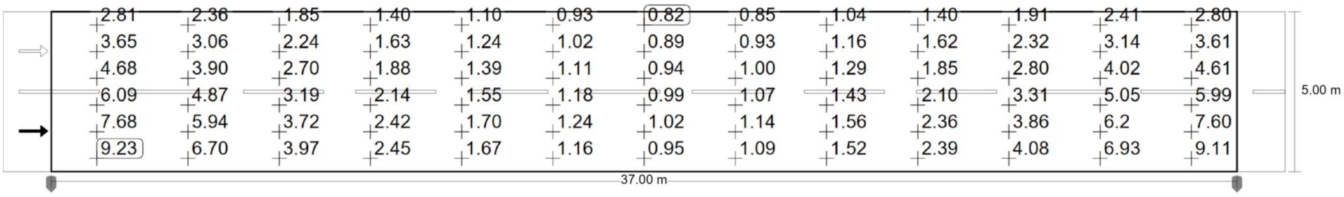
Roadway 1 (P6)

Results for valuation field

	Symbol	Calculated	Target	Check
Roadway 1 (P6)	E_{av}	2.78 lx	[2.00 - 3.00] lx	✓
	E_{min}	0.82 lx	≥ 0.40 lx	✓



Maintenance value, horizontal illuminance [lx] (Iso-illuminance curves)



Maintenance value, horizontal illuminance [lx] (Value grid)

m	1.423	4.269	7.115	9.962	12.808	15.654	18.500	21.346	24.192	27.038	29.885	32.731	35.577
4.583	2.81	2.36	1.85	1.40	1.10	0.93	0.82	0.85	1.04	1.40	1.91	2.41	2.80
3.750	3.65	3.06	2.24	1.63	1.24	1.02	0.89	0.93	1.16	1.62	2.32	3.14	3.61
2.917	4.68	3.90	2.70	1.88	1.39	1.11	0.94	1.00	1.29	1.85	2.80	4.02	4.61
2.083	6.09	4.87	3.19	2.14	1.55	1.18	0.99	1.07	1.43	2.10	3.31	5.05	5.99
1.250	7.68	5.94	3.72	2.42	1.70	1.24	1.02	1.14	1.56	2.36	3.86	6.20	7.60
0.417	9.23	6.70	3.97	2.45	1.67	1.16	0.95	1.09	1.52	2.39	4.08	6.93	9.11

Maintenance value, horizontal illuminance [lx] (Value chart)

	E_{av}	E_{min}	E_{max}	U_o (g_1)	g_2
Maintenance value, horizontal illuminance	2.78 lx	0.82 lx	9.23 lx	0.30	0.09